JianghuSect 2 (`JS2-v2.6.0`) — Deep-Dive Feature & Logic Audit

System #02 (Part 1 of 2): NPC Brain & Decision Architecture — Epistemic Knowledge Tiers, God-View Information Leaks, Domain Literacy & Social Knowledge Propagation

- **Target Codebase**: `Js2 new` (`SIM.PATCH_VERSION = '2.6.0'`, `SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting'`)
- **Target Readers**: Autonomous Coding Models (**GPT** & **Gemini** Systems Architects / AI Engineers)
- **Core Modules Audited**: `js/cognition.js`, `js/ai.js`, `js/cpeco.js`, `js/content.js`, `js/wild_resources.js`, `js/wildlife.js`, `js/external_market.js`, `js/bottleneck.js`, `js/visitors_clues.js`
- **Audit Scope (Part 1 of 2 — Epistemic & Perception Layer)**:
    1. **The 4-Tier Epistemic Knowledge Architecture** (Distinguishing *Universal Common Sense* vs. *Skill/Craft Literacy* vs. *Office/Rank Privileged Records* vs. *Private/Witnessed/Heard Knowledge*)
    2. **`[BRAIN-EPI-01]`** Sham `COG.knowledgeScope` & Half-Wired `COG.perceive` (`.slice(0, 24)` Library Truncation Bug & Unread Perception Arrays)
    3. **`[BRAIN-EPI-02]`** Omniscient Telepathy Leaks Across Private Inventories, Hidden Rival/Beast Power, Remote Closed-Door Mental States & Unscouted Wilderness
    4. **`[BRAIN-EPI-03]`** Missing Domain-Literacy Gating & Subjective Estimation Error (Calculator-Perfect Breakthrough Odds, Toxicity, Cursed Gear Appraisal & Zero-Skill Catalyst Refinement)
    5. **`[BRAIN-EPI-04]`** Disconnected Dual-Memory Stores (`mind.memories` vs. `cognition.memory`) & Zero Information Transfer During Social Interactions (`chat` / `teach` / `festival`)
    6. **High-Performance Epistemic Belief Schema (`<= 0.15 ms/tick` Budget for 80 NPCs)**
- **Companion Document (Part 2 of 2 — Next Pass)**: Action-Space Orphans (*game systems NPCs never use or weight: Pagoda Floors 10–43, 4th Sanctum, Micro-Locations, Apex Trials, Voluntary Vows, Rune Socketing, Recipe Copying*), Utility Scoring & Softmax Math Bugs (`leisure` mood sign error, 60-tick cache staleness, empty `study` loop), and Goal / Dead-Petition State-Machine Defects.

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0. Architectural Principle: Bounded Epistemic Rationality in an Emergent World

In a pure autonomous simulation where **NPCs drive 100% of world outcomes** and **the player cannot command NPCs**, the quality of emergent storytelling depends on **what each NPC knows, how they came to know it, and how accurately they can interpret it**:

1. **If an NPC does not know a system exists (Action Blindness)** -> That system is dead weight, no matter how richly designed.
2. **If every NPC knows everything in `W` instantaneously with 100% mathematical precision (God-View Omniscience)** -> Individual skills (`perceptiveness`, `MEDICINE`, `ALCHEMY`, `FORGING`, `SCRIPTURE`, `survival`, `readPeople`), offices (`spymaster`, `treasury`, `wildwarden`, `healer`), social conversations (`tea_garden`), scouts, and personality biases (`optimism`, `pride`, `caution`, `greed`) lose half their meaning because a Rank-0 illiterate novice has the exact same perfect information as the Sect Master!

The Canonical 4-Tier Epistemic Knowledge Matrix (Target Architecture)

Every piece of world data accessed by `AI` or `COG` must belong to one of four explicit epistemic tiers:

Tier	**Knowledge Category**	**Who Knows It & When**	**Examples in JianghuSect 2**
Tier 1	**Universal Common Sense** (ความรู้พื้นฐานที่ทุกคนรู้)	**All living NPCs immediately**, without needing special skills or hub visits	Own hunger/fatigue/pain; meal times & canteen location; sleeping in dorms; infirmary treats wounds; basic labeled sect pills (`BASIC_HP`, `BASIC_WOUND`, `BASIC_VIGOR`); current local weather/season; public sect policies & Festival announcements; who holds public offices

Tier	Knowledge Category	Who Knows It & When	Examples in JianghuSect 2
			(`master`, `deputy`, Hall Elders); existence of public sect buildings (`training_ground`, `market`, `plaza`, `mission_hall`).
Tier 2	**Bulletin & Hub-Synchronized Public Info** (ข้อมูลส่วนกลางที่ต้องแวะดูหรือรับฟังข่าว)	Known immediately to **relevant Hall Officers**, and learned by regular disciples **when visiting info hubs** (`scripture`, `market`, `main_hall`, `mission_hall`, `canteen`) or **hearing from peers** (`pair: chat/teach`)	Newly cataloged manuals in `W.sect.library` / `subLibrary` / `unlockedPaths`; current armory stock in `W.sect.gear`; active `shortageDirectives` & `officeOrders`; arrival of external `visitors`/`traders`; public `bountyBoard` & open Dimensional Gate announcements; newly enshrined `legacyTablets`.
Tier 3	**Domain & Craft Literacy** (ความรู้เฉพาะทางตามทักษะ สติปัญญา และประสบการณ์)	**All NPCs can see the object**, but **accuracy/depth scales with domain skill & intellect** (`ALCHEMY`, `MEDICINE`, `FORGING`, `ARRAY`, `BEAST_TAMING`, `SCRIPTURE`, `comprehension`,	True `breakthroughChance` vs. subjective overconfidence; subtle meridian `toxic` buildup (`<45`) & multi-phase pill side-effects; hidden cursed item drawbacks & Living Weapon / Catalyst / Rune potential; Rare

Tier	Knowledge Category	Who Knows It & When	Examples in JianghuSect 2
		`perceptiveness`, `readPeople`)	Beast Organ properties & Experiment tag synergies; gauging a stranger's/beast's concealed combat power across stage gaps; detecting insincere flattery or market overpricing.
Tier 4	**Privileged & Private Knowledge** (ความลับข้อมูลเฉพาะตำแหน่งและสิ่งที่ประสบเอง)	Strictly restricted to **the owner**, **direct witnesses**, **authorized officers** (`treasury`/`discipline`/`spymaster`/`master`), or **explicit confidants** (`spouse`/`mentor`/`friend` via social sharing)	Contents of *another* NPC's `n.inv` (`stones`, secret `recipes`, private `techs`, `organs`); exact `W.sect.stones` treasury balance & `auditTrail` embezzlement evidence; unscouted `wildResources` / `wildlife` spawns far outside sight range before a warden/gatherer spots them; another NPC's internal `heartDemon` or secret `demonPact` (unless symptomatic or confessed/diagnosed by `healer`); personal `knownClues` before archiving to `W.sect.clueArchive`.

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1. [BRAIN-EPI-01] Sham COG.knowledgeScope & Half-Wired COG.perceive (With .slice(0, 24) Library Blindness Bug)

1.1 Exact Code Anchors

- js/cognition.js:
  - COG.knowledgeScope (lines 250–264)
  - COG.isAwareOf & COG.visibleLibraryFor (lines 274–296)
  - COG.perceive (lines 299–316)
- js/ai.js: A.farm.score (line 559), A.mine.score (line 580), A.pray.score (line 665)
- js/cpeco.js: CPE.pickPurchase (line 249)

1.2 Root-Cause Code & Logic Defects

Defect 1A — COG.knowledgeScope Has Zero Runtime Callers

cognition.js:250–264 defines COG.knowledgeScope(W, n) returning { canSeeFullTreasury, canSeeAuditEvidence, canSeeApplicantCA, canSeeHiddenPA, knownZones }.

- **Actual Code Reality**: COG.knowledgeScope is **never called anywhere in cognition.js, ai.js, cpeco.js, or sect.js**!
- Consequently:
  1. CPE.pickPurchase (cpeco.js:250) lets every Rank-1 disciple read exact W.sect.stones * 0.25 from the Sect Treasury vault when pricing stone_grant.
  2. A.mine.score (ai.js:580) lets every Rank-0 initiate read the Forge/Alchemy halls' internal W.sect.want dictionary directly.
  3. COG.prepare (cognition.js:587) reads raw W.sect.shortageDirectives directly instead of n.cognition.knowledge.knownShortageIds.

Defect 1B — 4 out of 5 Arrays Populated by COG.perceive Are Dead Write-Only State

Every 45–240 ticks, COG.perceive(W, n) allocates and populates five arrays on n.cognition.knowledge:

2. visibleTechIds — partially used (with a severe bug below!)

3. `visibleGearIds` — **NEVER READ ANYWHERE** (`COG.bestGearUpgrade` at line 464 and `CPE.pickPurchase` at `cpeco.js:255` iterate raw `W.sect.gear` directly!)
4. `knownShortageIds` — **NEVER READ ANYWHERE**
5. `knownTabletIds` — **NEVER READ ANYWHERE** (`COG.isAwareOf(W, n, 'tablet')` at line 286 ignores `knownTabletIds` and unconditionally returns `W.sect.legacyTablets.length > 0`!)
6. `knownCatalysts` — **NEVER READ ANYWHERE** (`COG.isAwareOf(W, n, 'catalyst')` at line 289 unconditionally returns `Object.values(W.sect.catalysts).some(v => v > 0)`!)

Defect 1C — `.slice(0, 24)` Silent Truncation Bug Hides High-Tier Manuals in `COG.perceive` (`cognition.js:306–310`)

Inspect lines 306–310 of `js/cognition.js`:

```javascript
k.visibleTechIds = n.cult ? rawLib.filter(id => {
  const t = CULT.tech(id);
  return t && (t.reqTier || t.tier || 1) <= cap && (t.path === 'ANY' || t.path === n.cult.path || t.legendary);
}).slice(0, 24) : [];
```

- **Why This Breaks Progression in Mature Sects**:
  - `W.sect.library` is ordered chronologically (starter Tier-1/2 techniques + 12 Universal Arts `path === 'ANY'` + Multi-Path weapon arts sit at indices `0..23`).
  - Once a sect accumulates `> 24` compatible entries in `W.sect.library`, any newly donated Tier-5, Tier-6, or **Tier-7 Legendary Manual** sits at the tail (`index >= 24`).
  - Even when an Elder walks into the Scripture Pavilion (`nearInfoHub = true`), `.slice(0, 24)` **chops off everything past index 23**! For the next 180 ticks (`COG.isAwareOf` window), `k.visibleTechIds.includes(newLegendaryId)` returns **`false`** even right after `COG.perceive(W, elder, true)`!
  - And if `COG.visibleLibraryFor` is later tightened to rely on `k.visibleTechIds`, `.slice(0, 24)` would **permanently blind elders to all high-tier manuals**!

1.3 Remediation Blueprint

1. Sort filtered `visibleTechIds` by `(t.legendary ? 100 : 0) + (t.tier || 1) * 10` descending (or store a compact `{ syncedAtTick, librarySeq, gearSeq }` watermark instead of slicing arbitrary 16/24-element ID arrays!).

2. Use a **Watermark Timestamp Architecture (`lastHubSyncAt`, `lastMarketSyncAt`, `lastBoardSyncAt`)** on `n.cognition.epistemic`:
    - Any hall addition (`library`, `subLibrary`, `unlockedPaths`, `gear`, `catalysts`, `shortageDirectives`, `visitors`, `legacyTablets`) stamps `item.addedAt = W.t`.
    - An NPC is aware of an item if `item.addedAt <= ep.lastHubSyncAt` OR `item.actorId === n.id` OR `COG.role(W, n)` manages that hall OR `W.t - item.addedAt >= 720` (half-day word-of-mouth bulletin propagation inside the sect).
    - Visiting `canteen` (`eat`), `plaza`/`tea_garden` (`leisure`/`pair`), `market` (`shop`), `scripture` (`study`), or `main_hall` (`duty`) refreshes the corresponding watermark in **O(1) time and 0 array allocations**!

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2. `[BRAIN-EPI-02]` Omniscient Telepathy Leaks Across Private Bags, Hidden Power, Remote Mental States & Unscouted Wilderness

2.1 Exact Code Anchors

- `js/content.js`: `C.daily` private scroll trading (lines 494–500)
- `js/cognition.js`: `COG.forecast('huntRisk')` (line 343), `COG.prepare` `help_friend` & `rival` (lines 591, 638), `COG.huntValue` (line 814)
- `js/ai.js`: `AI.findSupportBlessTarget` (lines 621–648)
- `js/wild_resources.js` & `js/wildlife.js`: Global iteration of `W.wildResources.nodes` and `W.wildlife.beasts`

2.2 Root-Cause Code & Contextual Defects

Defect 2A — Telepathic Scanning of Other Disciples' Private Spirit Stones & Secret Scrolls (`content.js:494–500`)

In `content.js` (`C.daily`):

```javascript
const buyer = W.npcs.find(b => b !== holder && !b.dead && !b.away &&
  ((b.post === (r.craft === 'ALCHEMY' ? 'alchemy' : 'forge')) ||
(b.h.skills[r.craft] || 0) >= r.minMastery - 10) &&
  (b.inv.stones || 0) >= price * 1.5 && !(b.inv.recipes || []).includes(id));
```

- **Contextual Absurdity**:
    1. `holder` is a greedy/low-loyalty disciple who secretly pocketed a formula scroll (`shouldKeepFormula`) instead of turning it in to the Sect.
    2. Yet `holder` **telepathically knows every other NPC's exact private wallet (`b.inv.stones`) and private scroll pouch (`b.inv.recipes`)**, executes an instantaneous teleportation sale across the map without `pair` proximity, and **never risks offering an illicit stolen sect expedition scroll to a high-`integrity` Discipline/Hall Elder who would confiscate it or report them**!

Defect 2B — Exact Hidden Combat Power Telepathy (`cognition.js:343, 638, 814`)

- In `COG.prepare` (`rival` goal), `COG.forecast('huntRisk')`, and `COG.huntValue`, every NPC reads the target's **exact `CULT.power(target)` / `beast.power`**!
- **Why This Contradicts Xianxia Lore ("真人不露相 — True Masters Conceal Their Aura")**:
    - In Xianxia/Jianghu worlds, a Stage-2 Qi Refiner **cannot** accurately read the true combat power of a Stage-6 Soul Transformation elder, a `SHADOW_ASSASSIN` / `NINPO_SHADOW` stealth master, an `ILLUSION` cultivator, or a disguised `wanderingFigure`!
    - Likewise, a hunter with `survival = 5` and `BEAST_TAMING = 0` facing a never-before-seen `BEAST_V240` mutant should have high **epistemic uncertainty**, where a reckless/proud novice (`courage = 85, caution = 20, perceptiveness = 30`) misjudges the beast's power downward and gets ambushed, while a veteran `wildwarden` (`survival = 80, perceptiveness = 80`) or holder of `TR_BL_DIVINE_PUPIL` (เนตรสวรรค์สองตาดำ) reads its true threat accurately!

Defect 2C — Telepathic Detection of Closed-Door `heartDemon` & Unseen Wilderness Spawns

2. `AI.findSupportBlessTarget` (`ai.js:630`) checks `(o.h.state.heartDemon || 0) > 35` across all `W.npcs` regardless of distance or `o.hidden` (inside closed rooms). Internal Heart Demons are invisible mental turmoil unless `heartDemon >= 65` (visible erratic Qi deviation), or `o` is a close bond (`spouse`/`mentor`/`disciple`/`friend` who talked recently), or `n` has high `MEDICINE` / `SOUL_DIVINE` / `readPeople` and is nearby!
3. `gatherWildResource` (`wild_resources.js`) and `wildHunt` (`wildlife.js`) let disciples inside the sect walls target freshly spawned herbs, ores, and beasts at the far edges of the outer map **before any guard, scout, or forager has laid eyes on that zone**!

2.3 Remediation Blueprint

1. **Perceived Power Estimator (`COG.estimateTargetPower(W, observer, target)`)**:
    - Compute deterministic (zero-RNG, hashed on `(observer.id, target.id, W.day)`) observation error bounded by:
        - Stage gap `max(0, targetEqStage - obsEqStage)`: each stage above the observer increases uncertainty band by `±12%` unless observer has `TR_BL_DIVINE_PUPIL` (`PIERCE_ILLUSION` tag).
        - Observer acuity: `perceptiveness`, `senseAcuity`, and (`survival` + `BEAST_TAMING` for beasts; `combat` + `readPeople` for humans) shrink the error band toward `0%`.
        - Personality bias: `(courage + pride - caution * 1.2)` tilts unknown uncertainty toward underestimation (overconfidence) or overestimation (prudence)!
2. **Zone Reconnaissance State (`W.sect.scoutedZones[zoneId]`)**:
    - Outer wilderness zones maintain `lastScoutedAt` (updated whenever a `guard`, `wildwarden`, `forage`, `explore`, returning `expedition`, or `WANDERING_DAO` / aerial `QINGGONG` flyer passes near or patrols). Nodes/beasts inside unscouted stale zones are discovered organically by patrolling wardens/explorers rather than triggering psychic cross-map sprints!
3. **Physical Peer-to-Peer Scroll Barter (`pair` / `market`)**:
    - Move `C.tradeRecipe` out of the disembodied `C.daily` loop into a physical `market` / `pair` interaction where the seller only approaches plausible crafters (`alchemy`/`forge` workers or `wealthy_trader` peers), estimates whether they might afford it, and **checks `target.integrity` / `discipline` role** so shady sellers don't blindly hawk stolen expedition scrolls to the Chief Disciplinarian!

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3. `[BRAIN-EPI-03]` Missing Domain-Literacy Gating & Subjective Self-Assessment

3.1 Exact Code Anchors

- `js/ai.js`: `A.cultivate.start` (lines 366–372), `A.breakthrough.score` (lines 467–470)
- `js/bottleneck.js`: `BN.daily` & `CULT.inspectProgressionBottleneck` (`plan === 'wait'` vs `'push'`)
- `js/cpeco.js`: `CPE.pickPurchase` (lines 226–260)
- `js/cognition.js`: `COG.modifiers` catalyst check (line 705)

3.2 Root-Cause Code & Logic Defects

Defect 3A — Every Illiterate Novice Has a Built-In Pocket Calculator for Exact `breakthroughChance` & `toxic`

In `A.breakthrough.score` (`ai.js:467`): `const p = CULT.breakthroughChance(n, { env: 0.05, pill: n.cult.btBonus || 0 });` Every NPC—even one with `intellect.comprehension = 15`, `SCRIPTURE = 0`, `MEDICINE = 0`, `pride = 95`, and `caution = 15`—knows their **exact true mathematical `p`** (including hidden root modifiers, meridian integrity, and tribulation layers)!

- **Lost Emergent Drama**: In authentic Xianxia, arrogant/impatient cultivators with shallow Dao comprehension **misread their own foundation**, believing they have an 80% chance when their real chance is 48%, rushing into Heavenly Tribulation and suffering catastrophic Qi deviation—**unless** a wise Mentor (`mentor_disciple` / `guidedStudy`) or Healer examines their meridians and warns them (`n.cognition.epistemic.diagnosedBtChance`)!

Defect 3B — Zero-Skill Disciples Attempt Ancient Sanctum Catalyst Forge Refinement (`cognition.js:705`, `ai.js:518`)

Look at `cognition.js:705`: `if (id === 'forge' && (mc.hasRefineCand = !!(mc.safe && n.rank >= 2 && W.sect && W.sect.catalysts && ECO.findCatalystRefinementCandidate(W, n)))) push('ขัดเกลาศาสตรา T4–T6 ด้วยผลึกโบราณ', 0.52);` And in `ai.js:518` (`A.forge.can`): `hall === 'forge' && W.sect._hasCatalysts && ... && !!ECO.findCatalystRefinementCandidate(W, n)`

- **Notice what is NOT checked**: **`n.h.skills.FORGING`!!**
- Even a `FORGING = 0` musician or healer who happens to wear a T4 robe sees an ancient Sanctum Primordial Catalyst in the sect vault, walks up to the Blacksmith's Anvil (`forge`), and **hammers an irreplaceable Tier-5/6 Primordial Catalyst into their own robe by themselves** instead of entrusting it to the Forge Master or requiring `FORGING >= minMastery`!

Defect 3C — Omniscient Cursed / Complex Item & Multi-Phase Pill Evaluation

- NPCs with `FORGING = 0` and `SCRIPTURE = 0` immediately know all hidden properties of cursed equipment (`cursedHpDrain`, `sanityStrain`) and multi-phase pill aftermath debuffs (`aftermathStatus`) without appraisal.

3.3 Remediation Blueprint

1. **Subjective Breakthrough Confidence (`COG.perceivedBreakthroughChance(W, n, rawP)`)**:
   - Compute `skillClarity = clamp((E(n, 'intellect.comprehension') * 0.45 + (n.h.skills.SCRIPTURE || 0) * 0.30 + (n.h.skills.MEDICINE || 0) * 0.25) / 85, 0.15, 0.98)`.
   - If `n.cognition.epistemic?.mentorVerifiedStage === n.cult.stage` (set when mentored via `pair: teach` or `guidedStudy`!), `skillClarity = Math.max(skillClarity, 0.92)`!
   - Otherwise, `egoBias = ((E(n, 'personality.pride') - 50) * 0.0022 + (E(n, 'personality.ambition') - 50) * 0.0018 - (E(n, 'personality.caution') - 50) * 0.0025 + (n.h.state.heartDemon || 0) * 0.0025) * (1 - skillClarity)`.
   - `perceivedP = clamp(rawP + egoBias, 0.05, 0.98)`. Use `perceivedP` in `A.breakthrough.score` and `BN` wait/push decisions (while actual `CULT.breakthrough` rolls against true `rawP`!), creating organic, character-driven hubris, tragedy, and genuine value for Mentors!
2. **Gate Catalyst Refinement & Rune Socketing on `FORGING` / `ARRAY` Craft Literacy**:
   - Require `(n.h.skills.FORGING || 0) >= 35` (or `n.post === 'forge' || n.assist === 'forge'`) to personally execute `catalystRefine` at the anvil; non-smiths who hold T4+ gear and know a catalyst exists should **submit an `'equipment_refine'` work request / petition to the Forge Officer** rather than hammering ancient relics themselves with 0 Forging skill!

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4. `[BRAIN-EPI-04]` Disconnected Dual-Memory Stores & Zero Information Propagation in Social Interactions

4.1 Exact Code Anchors

- `js/ai.js`: `AI.mem` (lines 763–767), `AI.pairTick` (lines 808–835), `A.festival.tick` (line 715)
- `js/cognition.js`: `COG.remember` (lines 800–812), `COG.relInfo` (lines 828–848)

4.2 Root-Cause Code & Logic Defects

Defect 4A — `n.h.mind.memories` (`AI.mem`) Is 100% Write-Only Dead Weight to the Brain

Throughout `ai.js`, `economy.js`, `bottleneck.js`, and `cultivation.js`, major life events call `AI.mem(W, n, text, weight, whoId)`:

- Surviving or failing a breakthrough (`'ทะลวงขั้นล้มเหลว'`)
- Demonic Qi backlash during meditation (`'ปราณมารตีกลับขณะเร่งบำเพ็ญ'`)
- Rejecting a date invitation (`'ปฏิเสธคำชวน'`)
- Witnessing a sworn companion or mentor die (`'เสียชีวิต'`)
- Receiving a breakthrough pill from the sect **NONE of these entries in `n.h.mind.memories` are EVER read by `AI.decide`, `COG.prepare`, `COG.modifiers`, or `COMBAT`!** Meanwhile, `COG.remember` (`n.cognition.memory`) only records 8 narrow tags (`hunt_loss`, `hunt_win`, `received_help`, `gave_help`, `kept_commitment`, `broken_commitment`, `unfair_allocation`, `suspected_flattery`).
- **Result**: An NPC who nearly died of Qi backlash during `breakthrough` 9 days ago (`AI.mem('ทะลวงขั้นล้มเหลว', 1)`) has **zero cognitive trauma or caution modifier** in `COG.remember` when deciding whether to recklessly push another un-pilled breakthrough!

Defect 4B — Zero Epistemic Gossip / Knowledge Transfer During `pair` (`chat`, `teach`, `date`) & `festival`

Inspect `AI.pairTick` (`ai.js:820–833`):

- When `a` and `b` spend 30–100 minutes chatting at the Tea Garden, dating by the Spirit Pond, or attending a Sect Festival, `AI.pairTick` updates `r.a` / `r.t` / `mood`—**and exchanges ZERO knowledge!**
- Neither disciple learns:
  - New library manuals / shop arrivals / external traders that the other disciple knows (`lastHubSyncAt = Math.max(a.sync, b.sync)`).
  - Dangerous beast / zone warnings (`hunt_loss` in `north_peak` shared as a `heard_warning` memory at `0.6x` severity!).
  - Unarchived rumor leads or social reputations (`unfair_allocation` / `broken_commitment` gossip about corrupt officers when chatting with a trusted friend!).

4.3 Remediation Blueprint: Lightweight O(1) Social Epistemic Exchange (`COG.shareKnowledgeOnInteract(W, a, b, mode)`)

Hook `AI.pairEnd` (for `chat`, `teach`, `date`) and `A.festival.tick` (every 40 ticks) to call `COG.shareKnowledgeOnInteract(W, a, b, mode)`:

1. **Hub Bulletin Sync**: Both `a` and `b` advance their `epistemic.lastHubSyncAt = Math.max(epA.lastHubSyncAt, epB.lastHubSyncAt)`—so one scholar who visited `scripture` or `market` naturally spreads word of new manuals, shortages, and visiting merchants across the Tea Garden!
2. **Hazard & Scout Sharing**: Copy the fresher `hunt_loss`zone warning (at `0.65×` severity, tagged `viaGossip: true`) so cautious sectmates avoid a lethal beast zone after hearing a survivor's story!
3. **Mentor Meridian Diagnosis (`mode === 'teach'`)**: When `a` (`rank > b.rank` or higher `MEDICINE`/`SCRIPTURE`/stage) teaches `b`, stamp `b.cognition.epistemic.mentorVerifiedStage = b.cult?.stage` and `mentorVerifiedUntil = W.t + 14 * 1440`, eliminating `b`'s subjective `breakthroughChance` misjudgment and alerting `b` to hidden meridian `toxic`!
4. **Bridge Major `AI.mem` Trauma Events into `COG.remember`**: Record `bt_backlash` (increases preference for `consolidate` / pill preparation on next attempt), `bereavement_nemesis` (creates an autonomous `avenge_grudge` goal when `killerId` is known!), and `saved_in_combat` (`COMBAT.resolveRowBodyblock` -> `received_help` gratitude toward the vanguard who bodyblocked for them!).

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5. High-Performance Epistemic State Specification (`<= 0.15 ms/tick` Guarantee)

To ensure **zero performance regression** (keeping 80-NPC `SIM.tick` well below `3.5 ms`, near the current `~0.75 ms/tick`), GPT and Gemini **MUST NOT** clone arrays per tick or run O(N²) scans:

```javascript
// Stored on npc.cognition.epistemic (JSON-serializable, bounded size < 300 bytes/NPC):
npc.cognition.epistemic = {
  lastLibrarySyncAt: 0,      // W.t watermark when NPC last visited/heard library news
  lastMarketSyncAt: 0,       // W.t watermark for armory, catalysts, traders & shortages
```

```
  mentorVerifiedStage: 0,    // stage whose breakthrough readiness was diagnosed
by a senior
  mentorVerifiedUntil: 0,    // W.t expiry of mentor diagnosis
  knownDangerZones: {}       // { [zoneId]: { powerEst, until } } — max 6 keys
};
```

- **Complexity**: Checking `W.sect.itemAddedAt[id] <= ep.lastLibrarySyncAt` is **O(1)** integer comparison (`0` allocations, `< 0.001 ms`).
- **Determinism**: Subjective estimation noise (`estimateTargetPower`, `perceivedBreakthroughChance`) **must** use a pure deterministic hash `U.hash32(npc.id, targetId, Math.floor(W.day / 3))`—**never `W.rng`**—so calling `COG.explain(W, n)` or `AI.decide` preserves 100% RNG & serialization purity!

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6. Mandatory End-to-End Verification Suite Spec (For System #02 Part 1)

Add `test/v270_npc_brain_part1_epistemic_autonomy_suite.js` verifying:

1. **`test_library_above_24_manuals_not_truncated`**: Populate `W.sect.library` with 40 compatible manuals (Tier 1..7, Legendary at index 39). Have an Elder visit `scripture`; assert `COG.visibleLibraryFor(W, elder)` includes index 39 immediately.
2. **`test_gossip_propagates_hub_awareness_and_zone_warning`**: Add a new manual to `W.sect.library` (`addedAt = W.t`). Verify isolated Disciple B (`at mine`) is initially unaware while Scholar A (`at scripture`) is aware; run `AI.pairEnd(W, A, B, { type: 'chat' })`; assert Disciple B is now aware of the manual and inherits A's `hunt_loss` zone caution without visiting `scripture`.
3. **`test_subjective_breakthrough_hubris_vs_mentor_diagnosis`**: Create an arrogant, low-comprehension, `MEDICINE = 0` disciple with true `breakthroughChance = 0.42`. Verify `COG.perceivedBreakthroughChance` overestimates (`>= 0.54`), causing them to favor pushing; then run `COG.completeMentorLesson` with an Elder and verify `perceivedBreakthroughChance` converges within `±0.03` of `0.42`, switching their plan to `consolidate`/`wait`.
4. **`test_zero_forging_disciple_does_not_self_hammer_primordial_catalyst`**: Place `CAT_ASTRAL_ESSENCE_5` in `W.sect.catalysts`; verify a `FORGING = 0` Rank-2 disciple wearing T4 armor cannot directly run `catalystRefine` at the anvil, whereas a `FORGING >= 40` smith or Forge Officer refines it cleanly.

5. **`test_epistemic_evaluators_consume_zero_rng_and_zero_mutation`**: Assert `W.rng.getState()` and `SIM.serialize(W)` are byte-for-byte identical before and after calling `COG.estimateTargetPower`, `COG.perceivedBreakthroughChance`, and `COG.explain`.